

Beyond Nodal Damping: A Self-Energy Bridge for Inverter-Dominated Grids

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Abstract—Low-inertia screening often reduces a detailed dynamic model to an L, M, D object. That shortcut is only useful if damping can be represented by a fixed nodal coefficient. The assumption is weakest in inverter-rich grids, where damping may be carried by phase-locked loops, current controls, droop loops, exciters, governors, and stabilizers. A nodal- D screen can then point to the wrong margin-setting mode.

The proposed reduced object keeps the condensed controller block as a Schur self-energy, $\Sigma(s)$, rather than replacing it with a scalar damping term. The study will evaluate this bridge on IEEE 39-bus ANDES scenarios built by replacing synchronous generators with grid-following inverter-based resources over a penetration sweep. For each scenario, the protocol compares the full ANDES eigensolve, the self-energy poles extracted by contour integration, and the scalar L, M, D reduction used as a negative control. Time-domain transients will be used as a separate check on the modal predictions. The same reduced object labels critical modes as network-family or control-family, giving weak-node and weak-link screening a planning interpretation: some weak elements call for topology reinforcement, while others call for control retuning.

Index Terms—inverter-based resources, small-signal stability, damping margin, phase-locked loop, nonlinear eigenvalue problem, weak-node identification, power-system planning

I. INTRODUCTION

Low-inertia operation is usually discussed as an inertia problem. The damping problem is less tidy. In a synchronous-generator model, inertia and damping sit close to the electromechanical coordinates. In an IBR-rich model, damping can be supplied by PLLs, current loops, droop loops, exciters, governors, and stabilizers. Those states are not a scalar correction to the Laplacian.

That distinction matters for screening. A scalar bridge reduces the full model to a second-order object with damping in a fixed nodal coefficient. The proposed ANDES study treats that scalar bridge as the negative control. It asks when a reduced model that collapses controller dynamics to a constant tracks the wrong pole, and when the frequency-dependent self-energy is needed.

II. FORMULATION

The bridge starts from the linearized state matrix. Retained network/interface states are x_r ; condensed controller and auxiliary states are x_c :

$$\dot{x} = \begin{bmatrix} A_{rr} & A_{rc} \\ A_{cr} & A_{cc} \end{bmatrix} x, \quad x = \begin{bmatrix} x_r \\ x_c \end{bmatrix}.$$

At a trial pole s , the controller block is condensed through the resolvent, not replaced by a constant:

$$x_c = (sI - A_{cc})^{-1} A_{cr} x_r, \\ S_r(s) = sI - A_{rr} - A_{rc}(sI - A_{cc})^{-1} A_{cr}.$$

The same reduced object can be written as a second-order self-energy bridge:

$$T(s) = s^2 M + s \Sigma(s) + L, \\ \Sigma(s) = D_0 + C(sI - A_{cc})^{-1} B.$$

Two scalar diagnostics carry the engineering meaning:

$$\zeta(s) = -\frac{\Re(s)}{|s|}, \quad \pi_c = \frac{\|\hat{x}_c\|_2^2}{\|\hat{x}_r\|_2^2 + \|\hat{x}_c\|_2^2}.$$

Here ζ is the damping margin and π_c says whether a critical pole is network-family or control-family. ANDES poles enter only after this reduced extraction, for validation.

III. ANDES VALIDATION PROTOCOL

Evaluation will use the IEEE 39-bus system in ANDES. Starting from the packaged synchronous case, scenarios will replace selected synchronous generators with grid-following IBR models while preserving a solved operating point. The main sweep is IBR penetration. Secondary sweeps will vary dispatch, loading, and controller parameters such as PLL bandwidth and current-control gains.

For each scenario, ANDES will solve the power flow, form the full small-signal Jacobian, and compute the reference eigensolve. The same linearization will be condensed into $S_r(s)$ and $T(s)$; poles from the self-energy bridge will be extracted by contour integration. Validation will report pole-matching error, damping-margin error, and agreement on the family of the margin-setting pole. Transient checks will apply small disturbances such as load steps, set-point changes, and selected line outages, then compare the observed dominant frequency and damping with the eigenvalue predictions.

IV. WEAK NODES AND WEAK LINKS

With the controller self-energy retained, weak-node and weak-link analysis becomes a family classification. A compact local line score is the predicted damping loss

$$\eta_e = \max_{k \in \mathcal{K}} \{-\zeta_{k,we}, 0\}.$$

A network-family weak node points to topology reinforcement. A control-family weak node points to tuning.

Line reinforcement exposes a narrower case to be tested in the sweep. Some reinforcements may be control-resonant: they can pull a network mode toward a controller pole and lower damping rather than raising it. The audit will report how often this happens and whether it is tied to controller family, loading, or IBR location.

V. NEXT STEPS

Next steps are to run the penetration and controller-parameter sweeps, then compare the weak-element ranking against Fiedler participation, impedance/SCR, power-flow loading, and damping-participation baselines. The audit should separate three outcomes: topology reinforcement that improves damping, control retuning that moves the controller pole, and line actions that are rejected because they reduce distance to a control pole.